ProfileMetaNanni — the actual workflow

Phases A and B: implemented and tested · Phase C: designed, optional, off by default

INPUT

samples fetched from NCBI (SRR accessions) and/or your own FASTQ files

sensitive data: human contaminating reads, company recipes and batch identifiers

▼

ON YOUR LAPTOP · ON YOUR SERVER · OR ON THE HPC CLUSTER

Throughout this part the data **never leaves the machine**. No cloud, no API.

PHASE A · preprocessing — IMPLEMENTED

Fetch the samples
SRA-toolkit
only if they come from NCBI

Clean the reads
fastp + FastQC

Remove the host
Bowtie2 · human T2T-CHM13 always
+ pig / cattle / chicken as chosen

Quality report
MultiQC

PHASE B · read-based profiling — who is in there — IMPLEMENTED

Who is there
MetaPhlAn 4 and/or
Kraken2 + Bracken

Taxa × samples matrices
one row per species
one column per sample

Cross-tool comparison
only when tool = "both"
shared / one-only / Jaccard

Diversity
CLR + Aitchison + PCA
Kruskal-Wallis + Dunn

RESULTS — abundance tables · Excel report · PCA and diversity plots

these are small files: a few hundred kilobytes. They are all Phase C needs.

▼

PHASE C · taxa dossier with Biomni — DESIGNED

OPTIONAL · off by default · runs locally only, never submitted to the cluster

Take the NAMES only
species above threshold
no sample names, no SRR
no abundances, no FASTQ

Ask your permission
tells you who receives them
and that **it costs you money**
you can always say no

Biomni really searches
A1 agent + PubMed
role in fermented meat
+ expected functions

Re-check against NCBI
every citation is verified
if it does not match, **it is dropped**

Paid model — needs YOUR own API key
Claude (Anthropic) · GPT (OpenAI) · Gemini · Groq · Azure · Bedrock
reasons better · **costs real money** · species names leave the machine
if you have no key, the tool walks you through creating one

Local model — free
Ollama, running on your own computer
zero cost · **nothing leaves the machine** · no key to create
but it makes more mistakes here, and needs a powerful computer

▼

TAXA DOSSIER

for each species: its role in fermented meat, the functions to expect, with verified citations

every line is labelled **known** (demonstrated) · **expected** (inferred) · **hypothesis** (no evidence)

it exists to decide whether, and on which genomes, real functional annotation is worth the compute

⚠️ Honesty — what actually leaves your machine

- **Phase A and Phase B:** nothing leaves. Ever. No cloud.
- **Phase C with a local model:** nothing leaves there either.
- **Phase C with a paid model: species names only**, and only after you said yes.
- **The cost cannot be predicted up front:** it starts at 5 species so you can measure it, then you decide.
- **The dossier is written by an LLM, not by an expert:** that is stated at the top of it, and citations are re-checked against NCBI.

The LLM is the guide: it interviews the user, runs the scripts and interprets the results.
***It never improvises commands**: it only executes the tested scripts shipped in the repository (`bin/` and `steps/`).*